

Section-A (25 Marks)

1. a. What factors should be considered while formulating an irrigation scheme.[5]

b. What are the modes of irrigation? What are the advantages and disadvantages of each irrigation methods? [5]

Answer:

Irrigation is the process of artificial application of water to the crops for the better productivity. An irrigation scheme should be planned to provide an adequate, reliable, and economical water supply while ensuring sustainable use of land and water resources. The factors that should be considered while formulating irrigation scheme are as follows:

- i. Water availability:
 - The quantity and reliability of water from rivers, reservoirs, lakes, or groundwater are assessed.
 - The seasonal variations in water supply are studied.
- ii. Soil characteristics:
 - Soil type, texture, permeability, infiltration rate, and water-holding capacity must be properly studied and analyzed.
 - The suitability of soil for irrigation and drainage must also be determined.
- iii. Climate:
 - Rainfall pattern and distribution must be determined.
 - Temperature, humidity, wind, and evapotranspiration are also collected.
 - Crop water requirements.
- iv. Topography:
 - Land slope and terrain.
 - Suitability for gravity irrigation or the need for pumping.
 - Ease of canal alignment.
- v. Crop water requirement and pattern:
 - Type of crops to be grown.
 - Cropping intensity and growing seasons.
 - Water requirement of each crop
- vi. Economic feasibility:
 - Construction, operation, and maintenance costs.
 - Expected agricultural benefits and cost–benefit analysis.

b.

There are several methods of irrigation adopted based on the suitability of local condition such as soil characteristics, topographic conditions, crop rotation, available water flow and others. Few methods of irrigation are briefly described along with its advantages, limitations and suitability.

i. Surface irrigation:

- Surface irrigation is the most common type of the irrigation where the water is applied on the field in varied quantities and at different times.
- In this method, stream of water is diverted from the head of a field into the furrows or borders and allowing it to flow downward.
- Advantages:

- a. It allows to store the required amount of water in the capillary zone of the soil for supply of root zone of plants.
- b. It allows the use of machinery for land preparation cultivation and harvesting.
- Limitations:
 - a. Greater loss of water by surface runoff and deep percolation.
 - b. Inferior quality crops with a low yield are produced from the surface irrigation.
- Suitability:
 - Surface irrigation is most suitable for the land with the gentle slopes like terai region of Nepal.
 - This method is suitable for row and field crops.



ii. Sprinkler irrigation:

- Sprinkler irrigation is an improvement over conventional surface irrigation method.
- This method simulates natural rainfall to spread water in the form of rain uniformly over the land surface as needed at uniform pattern and at less rate than the infiltration rate of the soil to avoid the surface runoff.
- Advantages:
 - Provides protection to small plants from wind damage and soil from blowing.
 - Sprinklers provides protection against freezing injury to crops.
- Disadvantages:
 - Initial cost of installation is relatively high and regular maintenance must be done due to frequent clogging of the system.
 - This method is not so suitable at plain land as pressure head and sediment free water is required.
- Suitability:
 - Sprinkler irrigation is highly suitable for sandy soils, rolling and uneven terrain.
 - It is suitable for closely spaced crops like cereals and fodders.



iii. Drip irrigation:

- Drip irrigation is the method of applying water directly to the plants through a number of low flow rate outlet called emitter or drippers, placed at short interval along with small tubing.
- The network of drippers supply water and fertilizer to the plant roots at the rate of 1 to 2 liter per hour.

- **Advantages:**
 - Reduced percolation and evaporation losses resulting in economical use of water.
 - Fertilizer and nutrient losses minimized due to localized application and reduced leaching.
- **Disadvantages:**
 - Frequent clogging of the dripper requires constant supervision.
 - Not suitable for very closely seeded crops such as carrots, radishes etc.
- **Suitability:**
 - This method of irrigation is highly suitable for arid regions and sloppy terrain where the water is scarce.
 - The method is ideal for row crops, orchard and vineyards.



2. a. What do you understand by the duty of water at the headworks of an irrigation canal and at outlet to the field? [5]
- b. What measure would you suggest to improve “duty” in an existing canal system. [5]

Answer:

Duty of water is defined as the number of hectares that can be irrigated by constant supply of water at the rate of 1 cumecs throughout the base period. The duty of water is not constant but it varies with various factors like soil conditions method of flowing method of application of water.

In open channel flow, the relationship between duty (D) and delta (Δ) is given by:

$$\Delta = \frac{8.64 \times B}{D}$$

or

$$D = \frac{8.64 \times B}{\Delta}$$

Where:

Δ (**Delta**) = Depth of water required by the crop during its base period (m)

D = Duty of water (hectares/cumec)

B = Base period of the crop (days)

The duty of water at the headworks of an irrigation canal:

- It is the area irrigated by 1 cumec of water diverted at the headworks of the irrigation canal during the crop's base period.
- Water losses occur while flowing through the main canal, distributaries, minors, and watercourses due to: Seepage, evaporation, leakage, unauthorized withdrawals.

Hence, head duty is smaller because more water is needed at the headworks to compensate for these losses.

The duty of water at outlet to the field:

- It is the area irrigated by 1 cumec of water delivered at the outlet/ field channel during the crop's base period.
- Since the water has already passed through the canal system, the canal conveyance losses have already occurred.
- Therefore, field duty is greater than head duty.

b. We can improve the duty of water by increasing the area irrigated by the same quantity of water by efficient water management and reducing losses.

The measures to improve duty in an existing canal system:

i. **Line the Canals**

- Canals can be lined with concrete, brick, or other suitable materials to reduce seepage losses.

ii. **Proper Canal Maintenance**

- Silt, weeds, and other obstructions can be removed regularly to improve water flow and reduce losses.

iii. **Reduce Water Losses**

- Leaks are repaired and unauthorized withdrawals from canals are prevented.

iv. **Improve Water Distribution**

- Equitable and timely water distribution must be ensured through proper canal operation and rotational water supply.

v. **Adopt Efficient Irrigation Methods**

- Sprinkler or drip irrigation is used where feasible to reduce application losses and increase water-use efficiency.

vi. **Improve Field Channels**

- Proper maintenance of field channels to minimize seepage and leakage before water reaches the fields.

3. **What are the cross drainage works required in an irrigation system? Define the salient features of each type of CD works? What should be the minimum foundation depth of aqueduct below river bed? State the formula used from the calculating of scour depth of CD structure. [10]**

Answer:

Cross-drainage structures are those structures provided where a road, railway, or canal crosses a natural stream, river, or drainage channel. In irrigation project, suitable structures are constructed at the crossing point for the easy flow of water in canal and drainage.

Types of cross drainage structures used in irrigation system:

i. **Aqueduct:**

- Aqueduct is a cross drainage structure constructed when the drainage flood level is below the bed of canal.
- Small drains may be taken under the canal and banks by a concrete or masonry barrel, whereas in the case of stream crossings it may be economical to flume the canal over the stream.
- Drainage water passes clearly below the canal.

Aqueduct

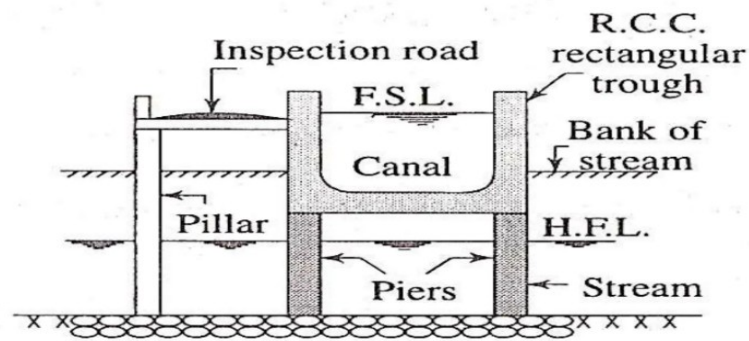


Fig: Aqueduct

ii. Siphon Aqueduct:

- In a hydraulic structure where the canal is taken over the drainage, but the drainage water cannot pass below the canal freely.
- It flows under siphonic action and designed for the total head loss through inverted siphon. So, it is known as siphon aqueduct.
- This structure is suitable when the bed level of canal is below the highest flood level of the drainage.

Siphon Aqueduct

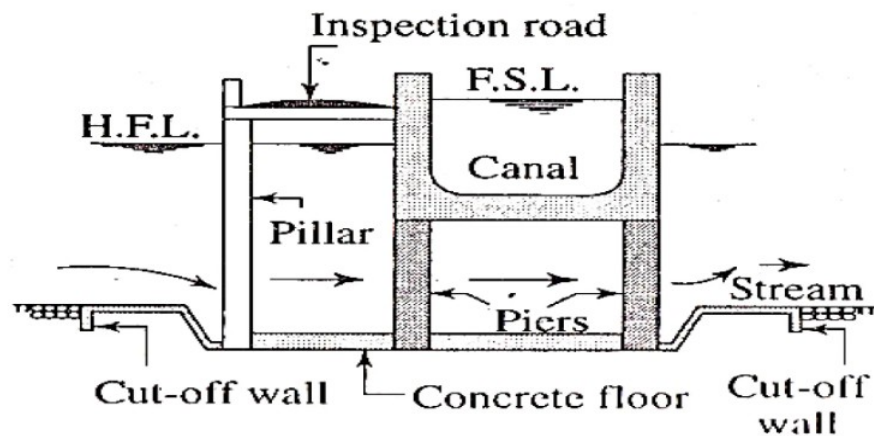


Fig: Siphon Aqueduct

iii. Super passage:

- The hydraulic structure in which the drainage is taken over the irrigation canal is known as super passage.
- The structure is suitable when the bed level of drainage is above the full supply level of the canal.
- The water of the canal passes clearly below the drainage.

Super Passage

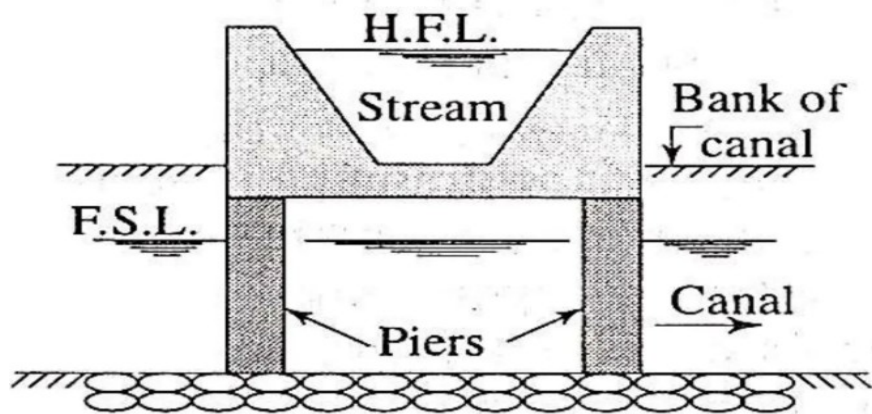


Fig: Super Passage

iv. Siphon super passage:

- The hydraulic structure in which the drainage is taken over the irrigation canal, but the canal water passes below the drainage under the siphonic action is known as canal siphon or siphon super passage.
- This structure is suitable when the bed level of drainage is below the full supply level of the canal.

Siphon Super Passage

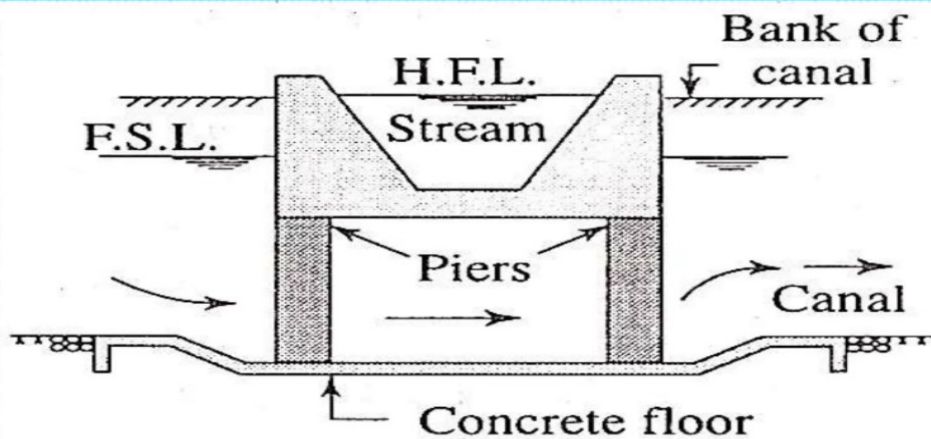


Fig: Siphon Super Passage

v. Level crossing:

- Level crossing facilities are provided when both the drain and the canal run at more or less the same level.
- This is more frequently used of drainage flows for short period of time like flash floods.
- The canal regulator regulates the flow with the drain regulator kept closed. Whenever the flash floods occur, the canal gate are closed.

Level Crossing

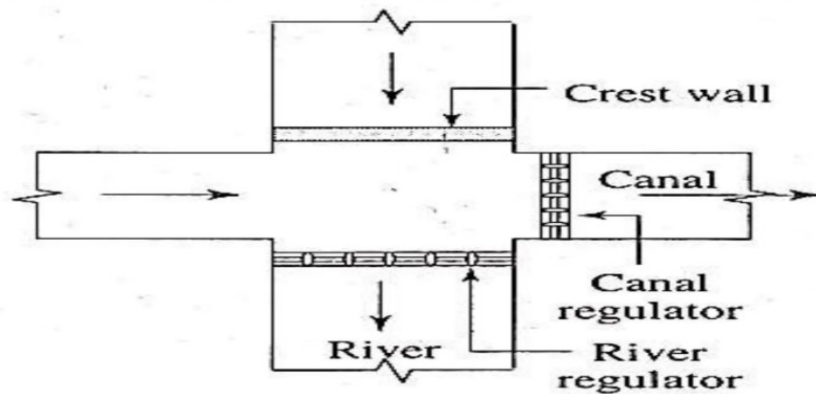


Fig: Level Crossing

vi. Canal inlet and outlets:

- When the drainage flow is small, it may be absorbed into the canal through inlets.
- The flow in the canal may be balance, if necessary, by providing suitable outlets or escapes.
- The inlet and outlets are provided with energy dissipators if necessary.

Inlet and Outlet

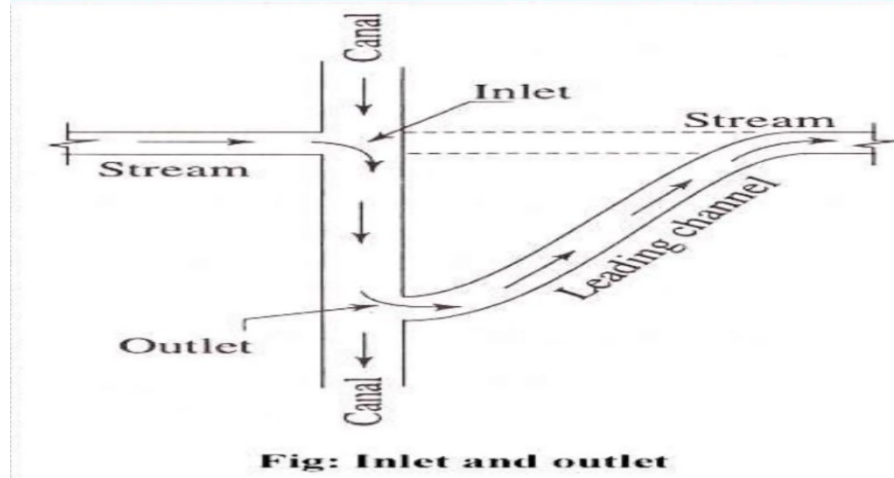


Fig: Inlet and outlet

The minimum foundation depth of aqueduct below river bed should be:

- At least 1.5 m below the calculated maximum scour depth for structures on alluvial rivers.
- In practice, the foundation level is generally kept at:

$$\text{Foundation Level} = \text{Scour Depth} + 1.5 \text{ m (minimum)}$$

- For rocky beds, the foundation should be anchored adequately into sound rock.

The scour depth of a CD structure is commonly calculated using Lacey's scour depth formula:

$$d = 0.473 \left(\frac{Q}{f} \right)^{1/3}$$

where:

d = Normal scour depth (m)

Q = Design discharge (m^3/s)

f = Lacey's silt factor

The silt factor is:

$$f = 1.76\sqrt{m}$$

where:

m = Mean particle size of bed material (mm)

4. a. What do you mean by Hydraulic Design Criteria of Cross Drainage Structures. Briefly mention the design procedure for determining uplift pressures in a floor of hydraulic structures. [7]

b. Write short notes on:

i. Scour depth

ii. Under Sluice

Answer:

Cross-drainage structures are those structures provided where a road, railway, or canal crosses a natural stream, river, or drainage channel. In irrigation project, suitable structures are constructed at the crossing point for the easy flow of water in canal and drainage.

Hydraulic Design Criteria of Cross Drainage Structures:

- i. The structure should safely pass the design flood discharge of the drainage and the design discharge of the canal without overtopping.
- ii. Hydraulic losses through the structure should be as small as possible to maintain the canal water level and discharge.
- iii. Flow velocity should be within permissible limits. Too high flow velocity causes scouring and too slow flow velocity causes sedimentation and silting.
- iv. The rise in upstream water level (afflux) due to the structure should be minimal to prevent flooding of adjacent lands.
- v. Proper inlet, outlet, and transition sections should be provided to minimize turbulence, eddies, and energy losses.
- vi. The structure should satisfy hydraulic requirements while remaining economical in construction and maintenance.
- vii. Cross drainage structure should remain stable against uplift pressure, sliding, overturning, seepage, and piping under all flow conditions.

Upward hydrostatic pressure exerted by seepage water beneath the floor of a hydraulic structure is known as uplift pressure. To ensure the safety of the floor of hydraulic structure against uplift and piping, uplift pressure must be determined.

Design procedure for determining uplift pressures in a floor of hydraulic structures:

- i. Upstream and downstream high flood level or water level is noted to calculate the total hydraulic head generating seepage.
- ii. Structural data such as length of the floor, depth and location of upstream and downstream cutoff, thickness of floor, foundation of soil properties are collected.
- iii. Residual head is determined at different points under the floor using Khosla's theory, Bligh's theory, Lane's weighted Creep theory.
- iv. Uplift pressure is calculated.

At any point,

$$p = \gamma_w \times h$$

where :

$$p = \text{uplift pressure (kN/m}^2\text{)}$$

$$\gamma_w = \text{unit weight of water } (\approx 9.81 \text{ kN/m}^3 \vee 10 \text{ kN/m}^3)$$

$$h = \text{residual pressure head (m)}$$

- v. Uplift pressure diagram is plotted beneath the floor. The maximum pressure occurs near the upstream end and decreases towards the downstream end.
- vi. Floor thickness of structure should be determined. It should be enough to resist uplift.
Check:

$$\text{Weight of floor} \geq \text{Uplift force} \times \text{Factor of Safety}$$

A factor of safety of about **1.2–1.5** is commonly adopted.

- vii. Increase the floor thickness if necessary. Provide upstream and downstream cutoffs. Install drainage galleries or pressure relief holes where applicable.

5. **Define exit gradient. How is exit gradient represented? Describe how a structure fails if exit gradient is not balanced. [10]**

Answer:

The hydraulic gradient of seepage flow at the downstream end of a hydraulic structure where seepage water emerges from beneath the impervious floor is known as exit gradient. It is the rate of loss of hydraulic head per unit length at the point where seepage exits. The exit gradient is a criterion for checking the safety of hydraulic structures against piping and undermining.

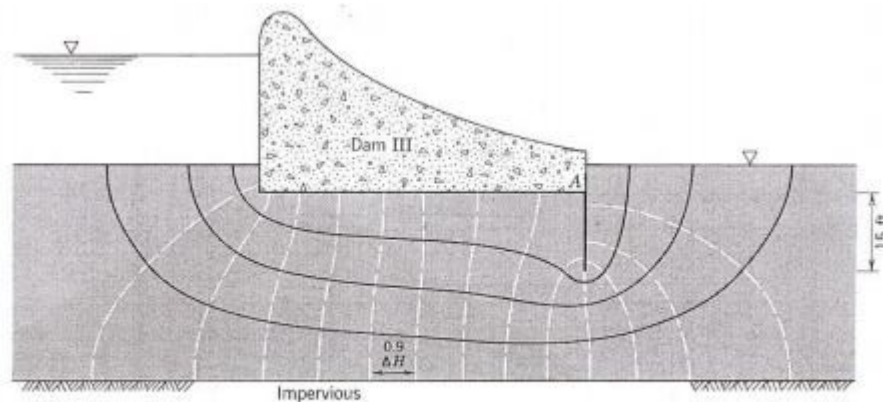


Fig. E18.3 Flow under three dams.

The exit gradient is represented as:

$$G_E = \frac{\Delta h}{\Delta l}$$

where :

$$G_E = \text{Exit gradient}$$

$$\Delta h = \text{Loss of hydraulic head near the exit (m)}$$

$\Delta l = \text{Length} - \text{which the head is lost (m)}$

According to Khosla's theory, the computed exit gradient must satisfy:

$$G_E \leq G_{Ec}$$

where:

$G_E = \text{Calculated exit gradient}$

$G_{Ec} = \text{Critical exit gradient of the foundation soil}$

Typical values of the critical exit gradient:

- Fine sand: **0.15–0.20**
- Coarse sand: **0.25–0.30**
- Gravel: **0.30–0.40**

If the actual exit gradient exceeds the critical exit gradient, the seepage force becomes greater than the submerged weight of the soil particles, which cause them to move. If the exit gradient is not balanced, structure fails by:

- i. **Boiling (Quick Condition)**
 - Soil particles lose contact with each other where the soil behaves like a liquid.
 - ii. **Piping**
 - Fine soil particles are carried away by seepage and underground tunnels (pipes) are formed.
 - iii. **Undermining of Foundation**
 - Continuous removal of soil creates cavities beneath the floor such that the foundation loses support.
 - iv. **Structural Failure**
 - The floor may crack, settle, or heave due to uplift and loss of support which ultimately cause the hydraulic structure to collapse.
6. a. What are the main causes of soil erosion and landslide in context to hill and mountainous areas of Nepal? Briefly mention their mitigation measures. [4]
b. Mention cause of flood in terai and inner valley. What are the methods of river training works with flood control generally adopted in flood plain? Also mention best solution of river control in alluvial plain in the terai.

Answer:

Soil erosion and landslides are the common natural calamities frequently occurring in hilly and mountainous area of Nepal. These calamities commonly occur in monsoon season costing huge loss of life and property. \

The main causes of soil erosion and landslide:

- i. High slope gradients increase runoff velocity and soil erosion, making slopes unstable.
- ii. Heavy rainfall saturates the soil, reduces its shear strength, and triggers landslides.
- iii. Highly weathered rocks, fractured rock masses, and loose colluvial deposits are prone to failure.
- iv. Removal of vegetation decreases root reinforcement and increases surface runoff and erosion.
- v. Improper hill cutting, steep cut slopes, and inadequate drainage destabilize slopes.
- vi. Concentrated surface runoff and seepage increase pore water pressure, leading to slope failure.
- vii. Cultivation on steep slopes without terracing or conservation measures accelerates soil loss.
- viii. Seismic activity weakens slopes and may trigger landslides in unstable terrain.
- ix. Undercutting of hill slopes by rivers removes toe support and induces landslides.
- x. Quarrying, mining, uncontrolled excavation, and unplanned urbanization disturb natural slope stability.

The mitigation measures are as follows:

- i. **Afforestation and Bio-engineering**
 - Plantation of vegetation, grasses, shrubs, and trees to stabilize slopes and reduce erosion.
- ii. **Proper Surface and Subsurface Drainage**
 - Construction of side drains, catch drains, cross-drains, and subsurface drains to safely remove water.
- iii. **Slope Protection Measures**
 - Provision of retaining walls, breast walls, gabion walls, stone pitching, rock bolting, shotcrete, and geotextiles where required.
- iv. **Terracing and Contour Farming**
 - Reduction of runoff velocity and soil erosion on agricultural land.
- v. **River Training Works**
 - Construction of check dams, spurs, revetments, and toe protection to prevent riverbank erosion.

b. The causes of flooding in the Terai and Inner Valley regions of Nepal are primarily linked to the geographical transition from the Himalayas to the plains, coupled with seasonal and climatic factors.

The main causes of floods are as follows:

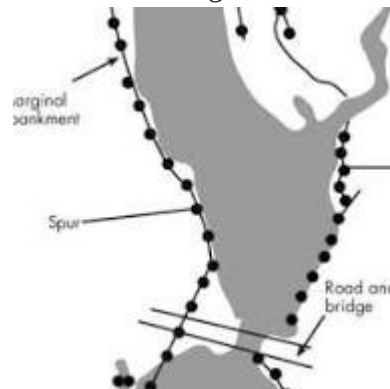
- i. Heavy monsoon rainfall in the Siwalik, Middle Hills, and Terai.
- ii. Flat terrain and low river gradient, which slows the flow of water.
- iii. Sediment deposition (aggradation) raises river beds, reducing their carrying capacity.
- iv. Riverbank erosion and embankment breaches during floods.
- v. Poor drainage and waterlogging due to inadequate drainage channels.
- vi. Deforestation and land degradation in upstream watersheds, increasing runoff.

In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting.

There are different methods of river training works. Some of them are described below:

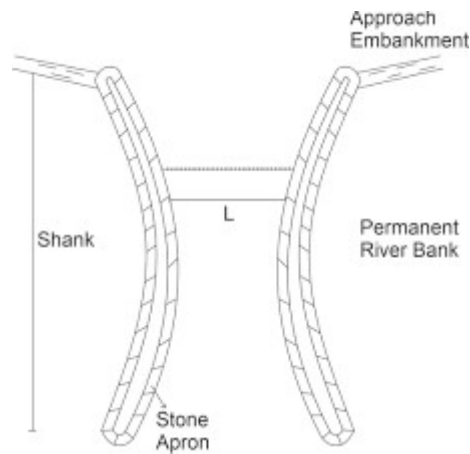
i. Marginal embankments/ levees:

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



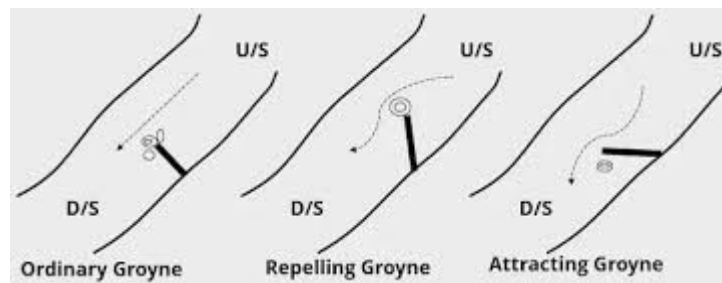
ii. Guide banks:

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



iii. Groynes or spurs:

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



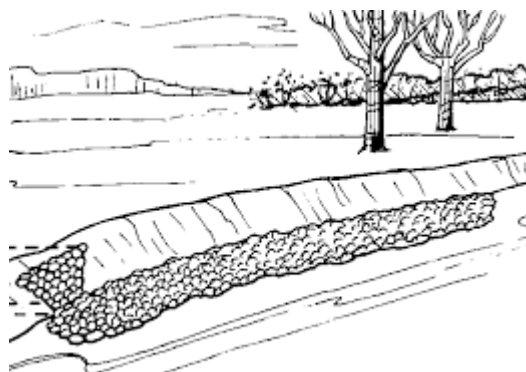
iv. Bank protection and Pitched banks:

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

Bank protection work can be classified as direct or indirect.

Direct protection works include work done on the bank itself such as vegetation cover, pitching

Indirect protection includes work constructed not directly on the banks but in front of them.



The best solution of river control in alluvial plain in the terai are the river training works such as construction of embankments, guide banks, spurs etc.

7. List the step-by-step logical design process for irrigation development of a medium scale irrigation project. [10]

Answer:

The step-by-step logical design process for irrigation development of a medium scale irrigation project are listed below:

- i. Reconnaissance survey:
 - Identify the project area and water source.
 - Assess topography, climate, and preliminary feasibility.
 - Identify possible canal alignment and command area.
- ii. Feasibility study:
 - Assess water availability (surface or groundwater).
 - Estimate irrigation demand and command area.
 - Conduct preliminary economic, technical, social, and environmental analyses.
 - Select the most suitable project alternative.
- iii. Detailed survey:
 - Topographic survey.
 - Hydrological and meteorological investigations.
 - Geological and geotechnical investigations.
 - Soil survey and land capability assessment.
 - Command area survey.
- iv. Hydraulic design:
 - Design headworks (weir/barrage/intake).
 - Design main canal, branch canals, distributaries, and minors.
 - Design canal sections, slopes, and velocities.
 - Design cross-drainage works, falls, regulators, escapes, and outlets.
- v. Structural design:
 - Design foundations and hydraulic structures.
 - Check stability against uplift, sliding, overturning, and seepage.
 - Prepare structural drawings and reinforcement details.
- vi. Environmental assessment:
 - Conduct IEE/EIA as required.
 - Prepare environmental management and mitigation plans.
 - Address land acquisition, resettlement, and social impacts.
- vii. Preparation of Detailed Project Report:
 - Prepare design drawings.
 - Estimate quantities and project cost.
 - Prepare construction specifications and implementation schedule.
- viii. Construction planning and implementation:
 - Procure contractors and materials.
 - Construct headworks, canals, and associated structures.
 - Supervise quality control and project progress.

8. Answer the following questions:

- a. i) What is hydrological cycle and what is its importance?
- ii) What is meant by rain gauge and what is their use?

iii) Explain intensity duration curves, infiltration and percolation.

b. Enumerate the different methods of measuring the discharge. How will you measure the discharge of i) a river ii) small stream iii) a canal?

Answer:

The hydrological cycle, also known as the water cycle, is the continuous circulation of water within the Earth's atmosphere system. This cycle has no beginning or end, occurring as a perpetual process that involves an exchange of energy leading to temperature changes.

The cycle is divided into two primary categories of components:

- **Transportational Components:** These include precipitation, evaporation, transpiration, infiltration, and runoff.
- **Storage Components:** These include soil moisture, groundwater storage, and storage on land (such as lakes and reservoirs).

Importance of hydrological cycle:

- i. The cycle continuously replenishes the land with fresh water, which is essential for human consumption and agriculture.
- ii. Knowledge of the cycle's runoff and flood patterns is critical for the safe design of bridges, culverts, weirs, and dams. For instance, scour depth for bridge foundations is determined based on design flood discharges obtained through hydrological analysis.
- iii. Efficient highway and airport drainage systems rely on understanding surface and subsurface water movement to prevent pavement softening, landslides, and hydroplaning.
- iv. The installed capacity of Run-of-River (ROR) plants and reservoir storage requirements are determined using cycle-driven data like Flow Duration Curves and mass inflow curves.
- v. The cycle is a primary indicator of climate change. Impacts such as intensified rainfall, prolonged dry periods, and the melting of glaciers (leading to Glacial Lake Outburst Floods) directly alter the natural water cycle in regions like Nepal.

Rain gauges are the instruments that are used for the measurement of precipitation. Precipitation is expressed in a vertical depth of water such as mm, inch, cm etc. they are classified into two types:

i. Recording types:

- These are rain gauges that give a permanent, automatic rainfall record in the form of a pen mounted on a clock driven chart.
- Weighing bucket type rain gauge, tipping bucket type rain gauge, float type of rain gauge are some of the recording type of rain gauge.

ii. Non-recording types:

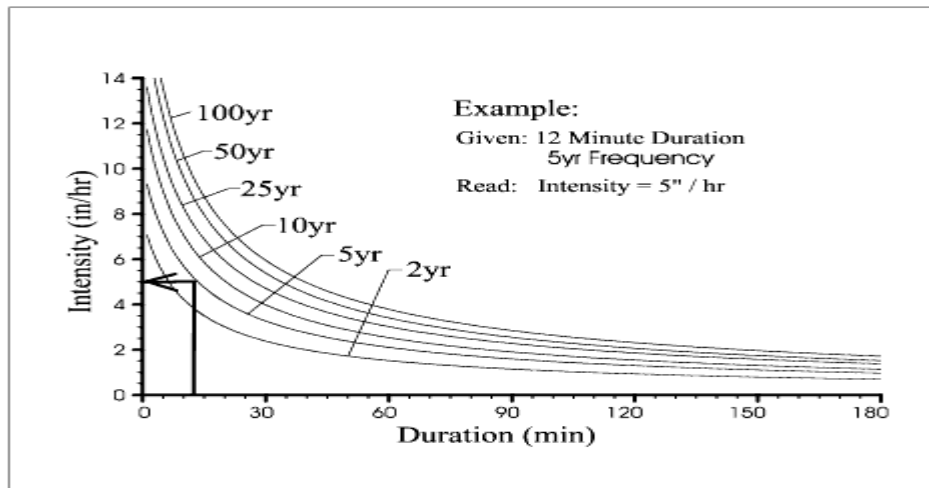
- These rain gauges do not record depth of rainfall, but only collect rainfall.
- Symon's rain gauge is the most common non-recording type of rain gauge.
- It gives total rainfall that has occurred at a period.

Use of rain gauges:

- i. Measures daily, monthly, seasonal, and annual rainfall.
- ii. Provides rainfall data for runoff estimation, flood forecasting, and watershed analysis.
- iii. Used in the design of dams, reservoirs, spillways, culverts, bridges, and drainage systems.
- iv. Helps estimate crop water requirements and schedule irrigation.
- v. Rainfall data are used to predict floods and issue early warnings.

Intensity Duration curves:

Intensity Duration curve is a graph that shows the relationship between rainfall intensity and the duration of a storm at a particular location. It helps in design of stormwater drainage systems, design of culverts, bridges, and highway drains.



Infiltration:

The process by which water on the ground surface enters and moves into the soil through its pores. It is an important part of the hydrologic cycle because it determines how much rainfall becomes groundwater recharge and how much becomes surface runoff.

The infiltration rate is the quantity of water entering the soil per unit area per unit time.

$$\text{Infiltration Rate} = \frac{\text{Volume of water infiltrated}}{\text{Area} \times \text{Time}}$$

Units: mm/h or cm/h

Percolation:

The downward movement of water through the soil under the influence of gravity after the water has infiltrated into the soil is called percolation. The percolation rate is the rate at which water moves downward through the soil.

Units: mm/h, cm/h, or mm/day.

The different methods of measuring the discharge are:

- i. Area-velocity method:
 - In this method, the discharge estimation is done by considering a cross section to be divided into a large number of subsections by verticals.
 - The average velocity in these subsections is measured by current meter or floats.
 - The accuracy of a discharge estimation increases with number of subsections used.
- ii. Current meter method:
 - The most commonly used instrument in hygrometry to measure the velocity at a point in the flow cross section is the current meter.
 - It consists of a rotating element which rotates due to reaction of the stream current with an angular velocity proportional to stream velocity.
- iii. Float method:
 - A floating object on the surface of a stream when time can yield the surface velocity.
 - Floats must be easily visible and leak proof.
 - Generally mean velocity is obtained by multiplying the observed surface velocity by a reduction coefficient.
- iv. Stage-discharge (rating curve) method:
 - The graph representing relation between the stage and discharge is called rating curve.

- If G- Θ relationship for a gauge station is constant, the control is said to be permanent. If it changes with time, it is called shifting control.

Measurement of discharges can be done by:

- i) River:** Measure discharge by the current meter (area–velocity) method.
- ii) Small Stream:** Measure discharge by the float method or by using a V-notch/rectangular weir.
- iii) Canal:** Measure discharge by the current meter method, Venturi flume, or weirs.

9. **a. What are the advantages and disadvantages of ground water irrigation is compared to surface canal irrigation?**

b. A 30 cm diameter well penetrates 25 m below the static water table. After 24 hours of pumping at 5400 liters per minute, the water level in a test well at 90 m is lowered by 0.53 m and well 30 m away, the drawdown is 1.11 m. Find out the transmissibility of the aquifer. [10]

Answer:

Shallow or deep wells, infiltration galleries etc. are used to tap groundwater from aquifers for irrigation uses, whereas surface canal irrigation supplies water through canals from rivers, reservoirs, or barrages.

Advantages of ground water irrigation is compared to surface canal irrigation:

- i. **Reliable Water Supply**
 - Water is available on demand and is less affected by canal operation.
- ii. **Better Farmer Control**
 - Farmers can irrigate crops whenever needed, improving crop management.
- iii. **Higher Water Use Efficiency**
 - Lower conveyance losses occur in groundwater since water is pumped directly to fields.
- iv. **No Waterlogging**
 - Does not generally cause waterlogging if groundwater is used sustainably.
- v. **Less Land Acquisition**
 - No need for extensive canal networks, reducing land loss.

Disadvantages of ground water irrigation is compared to surface canal irrigation:

- i. **High Initial and Operating Cost**
 - Requires high initial investment in wells, pumps, and energy (diesel or electricity).
- ii. **Groundwater Depletion**
 - Excessive pumping lowers the groundwater table and may dry up wells.
- iii. **Energy Requirement**
 - Continuous power or fuel is needed to operate pumps.
- iv. **Limited Yield**
 - Water availability depends on aquifer capacity; not suitable where groundwater is scarce.

Given,

Diameter of pumping well = 30 cm
 Penetration below static water table = 25 m
 Pumping rate, $Q = 5400 \text{ L / min}$

Observation well 1:

$$r_1 = 30 \text{ m}$$

$$\text{Drawdown, } s_1 = 1.11 \text{ m}$$

Observation well 2:

$$r_2 = 90 \text{ m}$$

$$\text{Drawdown, } s_2 = 0.53 \text{ m}$$

$$Q = 5.4 \text{ m}^3 / \text{min}$$

$$= 5400 \text{ L} / \text{min}$$

$$= \frac{5.4}{60} = 0.09 \text{ m}^3 / \text{s}$$

For a confined aquifer,

$$T = \frac{Q}{2\pi(s_1 - s_2)} \ln\left(\frac{r_2}{r_1}\right)$$

where

$$s_1 - s_2 = 1.11 - 0.53 = 0.58 \text{ m}$$

and

$$\ln\left(\frac{90}{30}\right) = \ln(3) = 1.0986$$

Substitute the values:

$$T = \frac{0.09}{2\pi(0.58)} \times 1.0986 = 0.0271$$

$$T = 0.0271 \times 86400 = 2341 \text{ m}^2 / \text{day}$$

Hence,

$$T = 0.027 \text{ m}^2 / \text{s}$$

or

$$T \approx 2.34 \times 10^3 \text{ m}^2 / \text{day}$$